HW5

Alex Villagomez

2026-03-06

Contents

Problem 1

a)

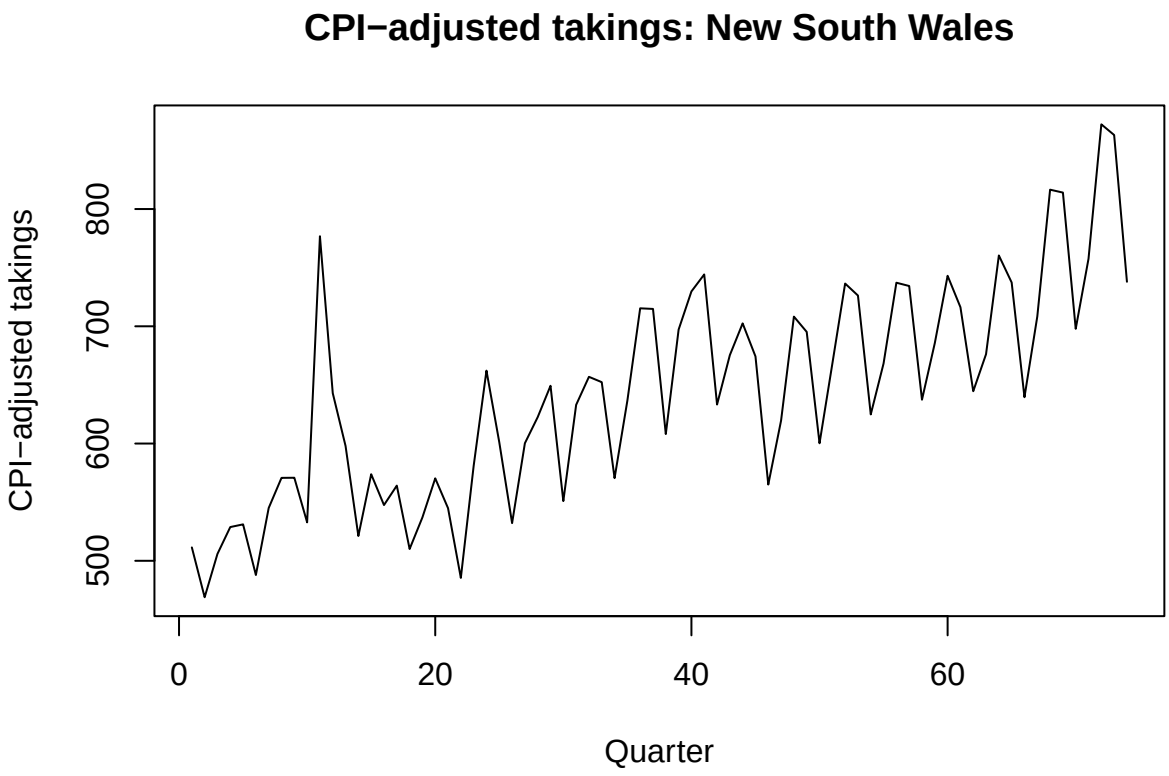
```
data(aus_accommodation)
aus_accommodation <- aus_accommodation %>%
  mutate(cpi_adj_takings = Takings / CPI * 100)
head(aus_accommodation)
```

```
## # A tsibble: 6 x 6 [1Q]
## # Key:       State [1]
##      Date State                           Takings Occupancy   CPI cpi_adj_takings
##     <qtr> <chr>                             <dbl>     <dbl> <dbl>            <dbl>
## 1 1998 Q1 Australian Capital Territory    24.3        65  67                36.2
## 2 1998 Q2 Australian Capital Territory    22.3        59  67.4              33.1
## 3 1998 Q3 Australian Capital Territory    22.5        58  67.5              33.4
## 4 1998 Q4 Australian Capital Territory    24.4        59  67.8              36.0
## 5 1999 Q1 Australian Capital Territory    23.7        58  67.8              35.0
## 6 1999 Q2 Australian Capital Territory    25.4        61  68.1              37.3
```

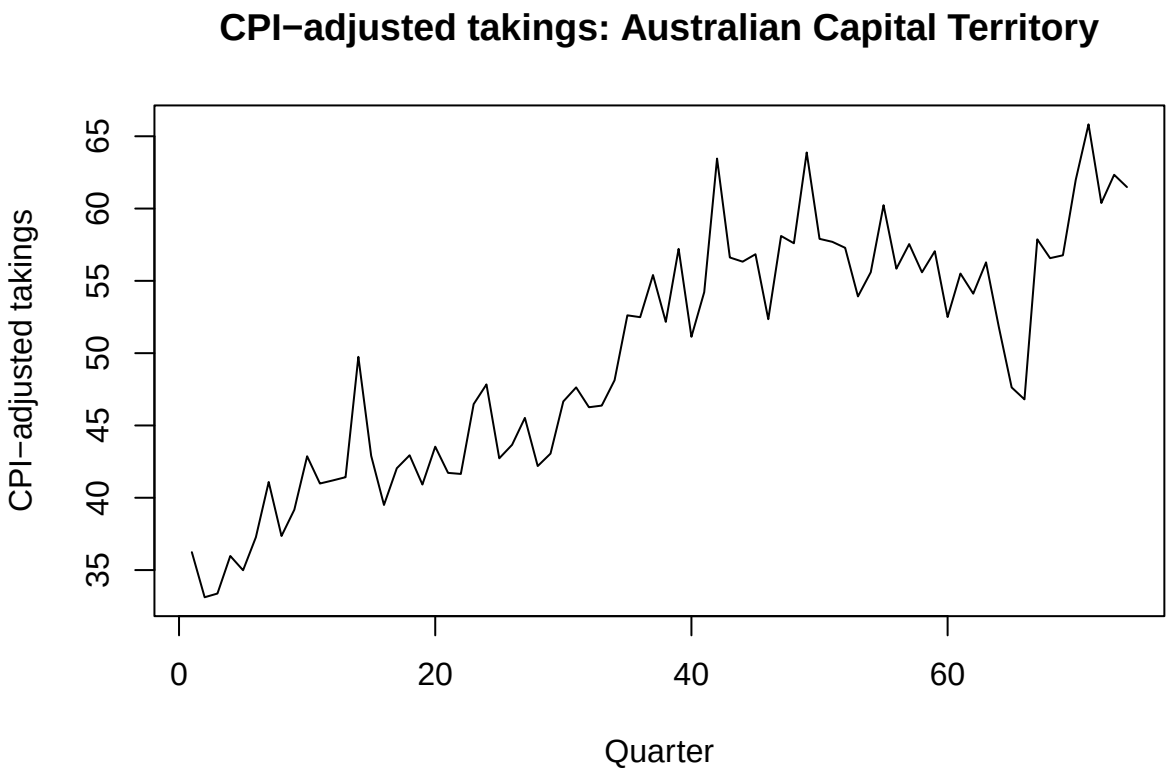
```
unique(aus_accommodation$State)
```

```
## [1] "Australian Capital Territory" "New South Wales"
## [3] "Northern Territory"           "Queensland"
## [5] "South Australia"              "Tasmania"
## [7] "Victoria"                     "Western Australia"
```

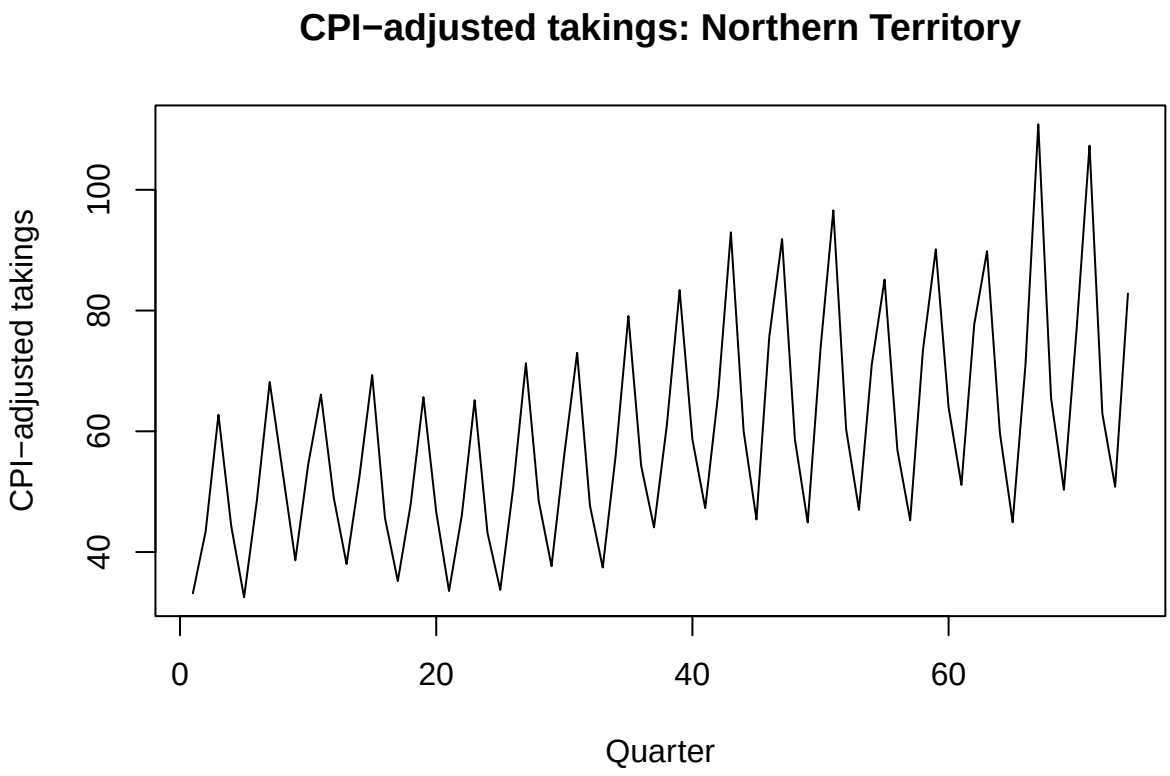
```
nsw <- aus_accommodation %>% filter(State == "New South Wales")
plot(nsw$cpi_adj_takings, type = "l", main = "CPI-adjusted takings: New South Wales", xlab = "Quarter",
```



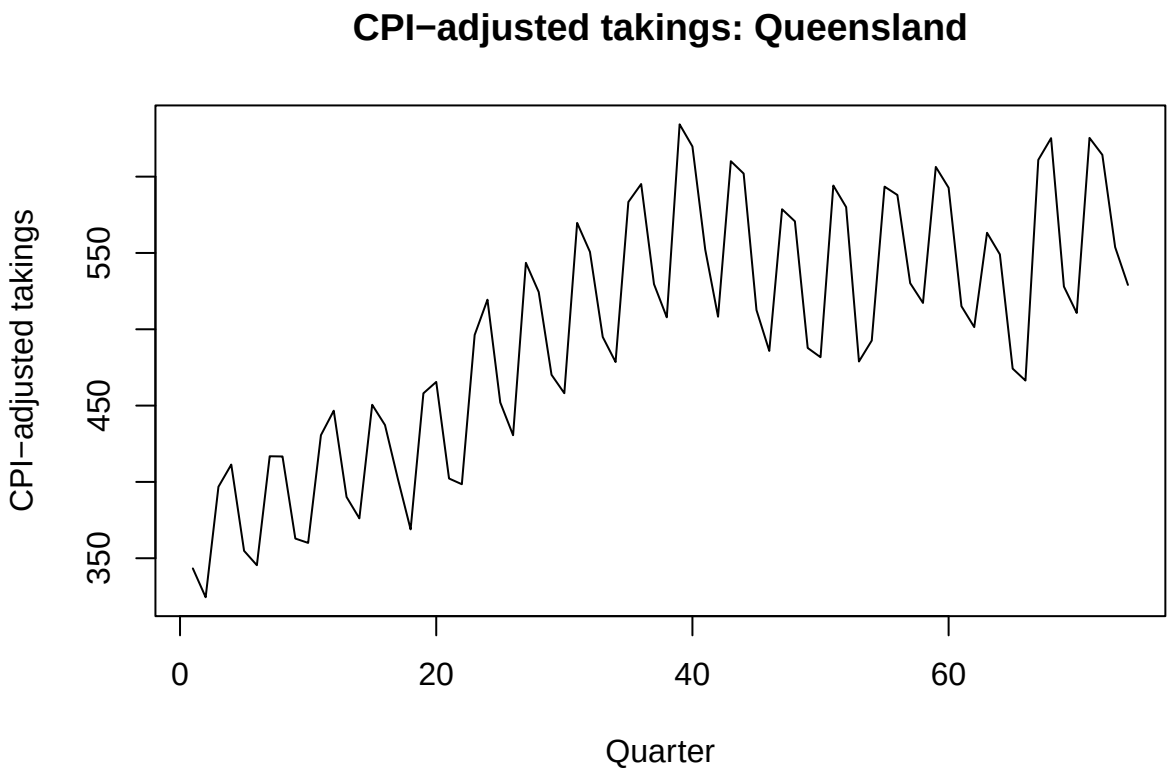

```
act <- aus_accommodation %>% filter(State == "Australian Capital Territory")
plot(act$cpi_adj_takings, type = "l", main = "CPI-adjusted takings: Australian Capital Territory", xlab
```



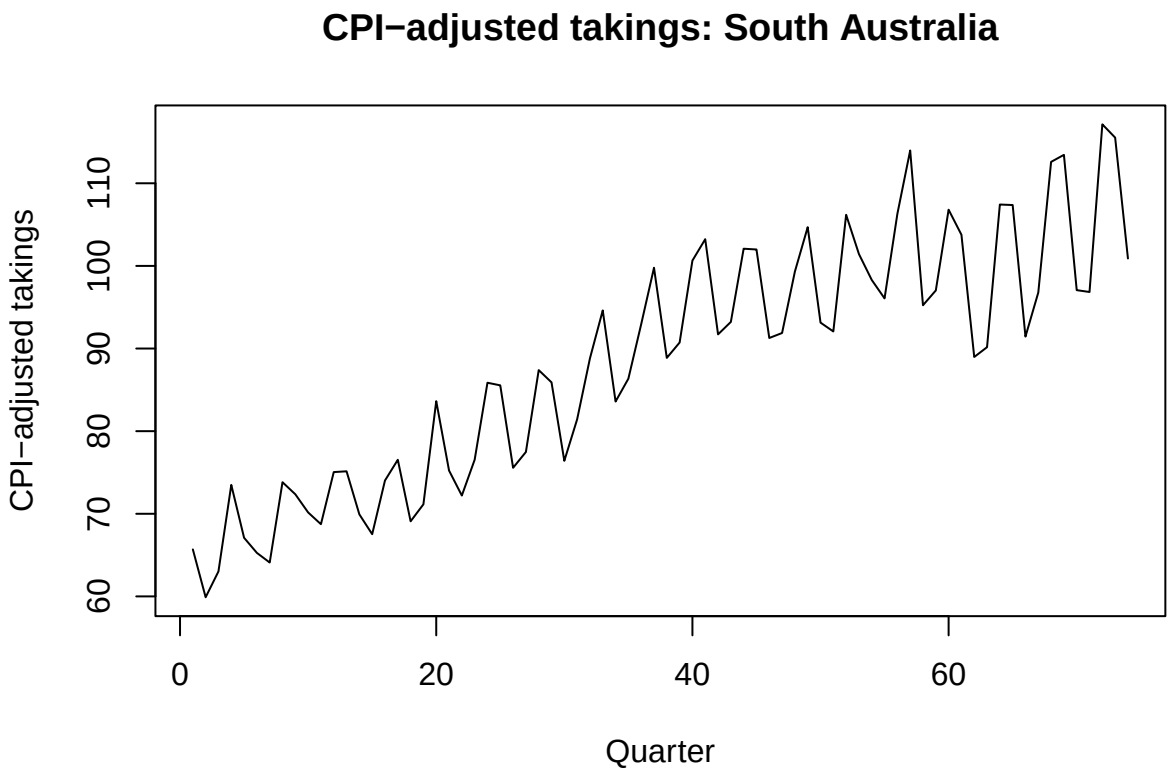

```
nt <- aus_accommodation %>% filter(State == "Northern Territory")
plot(nt$cpi_adj_takings, type = "l", main = "CPI-adjusted takings: Northern Territory", xlab = "Quarter
```



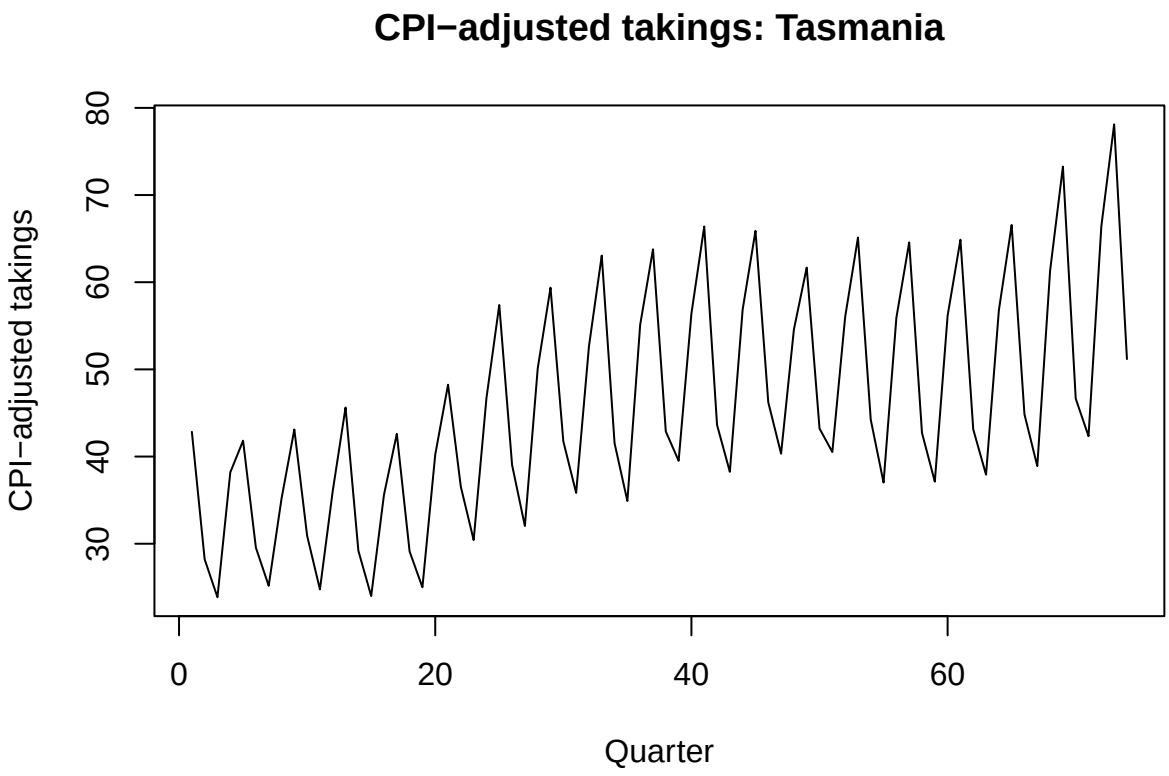

```
qld <- aus_accommodation %>% filter(State == "Queensland")
plot(qld$cpi_adj_takings, type = "l", main = "CPI-adjusted takings: Queensland", xlab = "Quarter", ylab
```



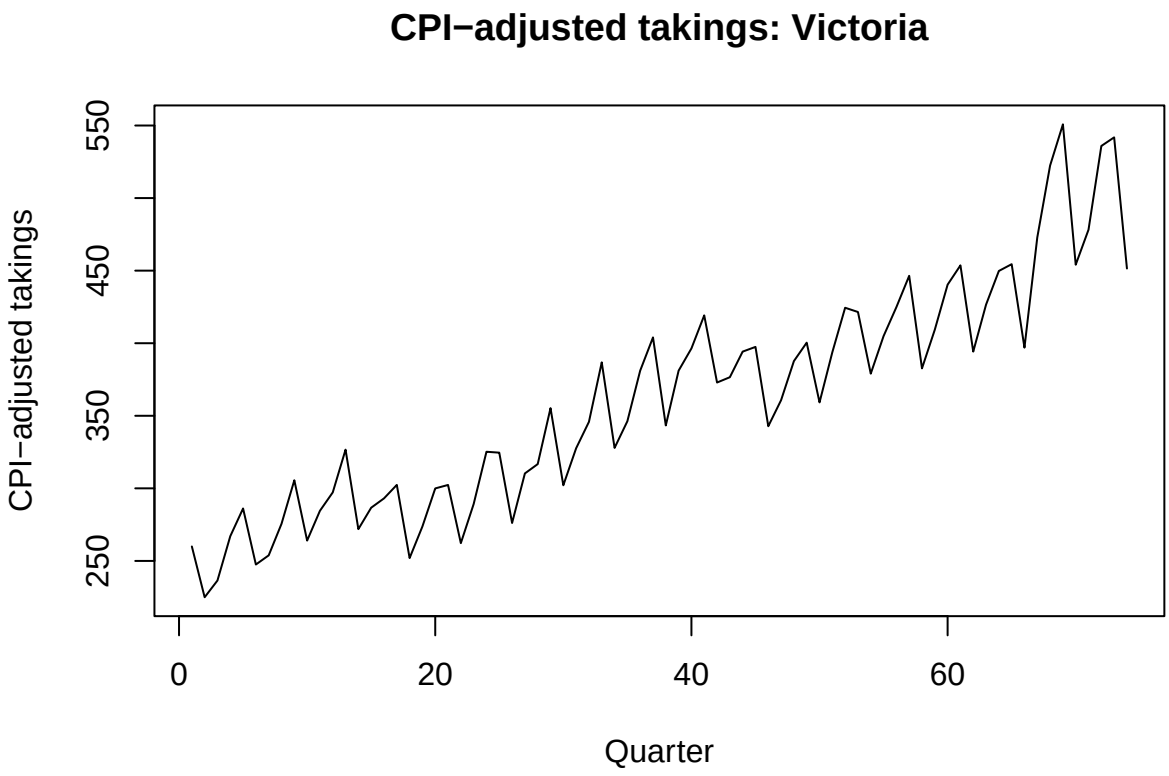

```
sa <- aus_accommodation %>% filter(State == "South Australia")
plot(sa$cpi_adj_takings, type = "l", main = "CPI-adjusted takings: South Australia", xlab = "Quarter", y
```



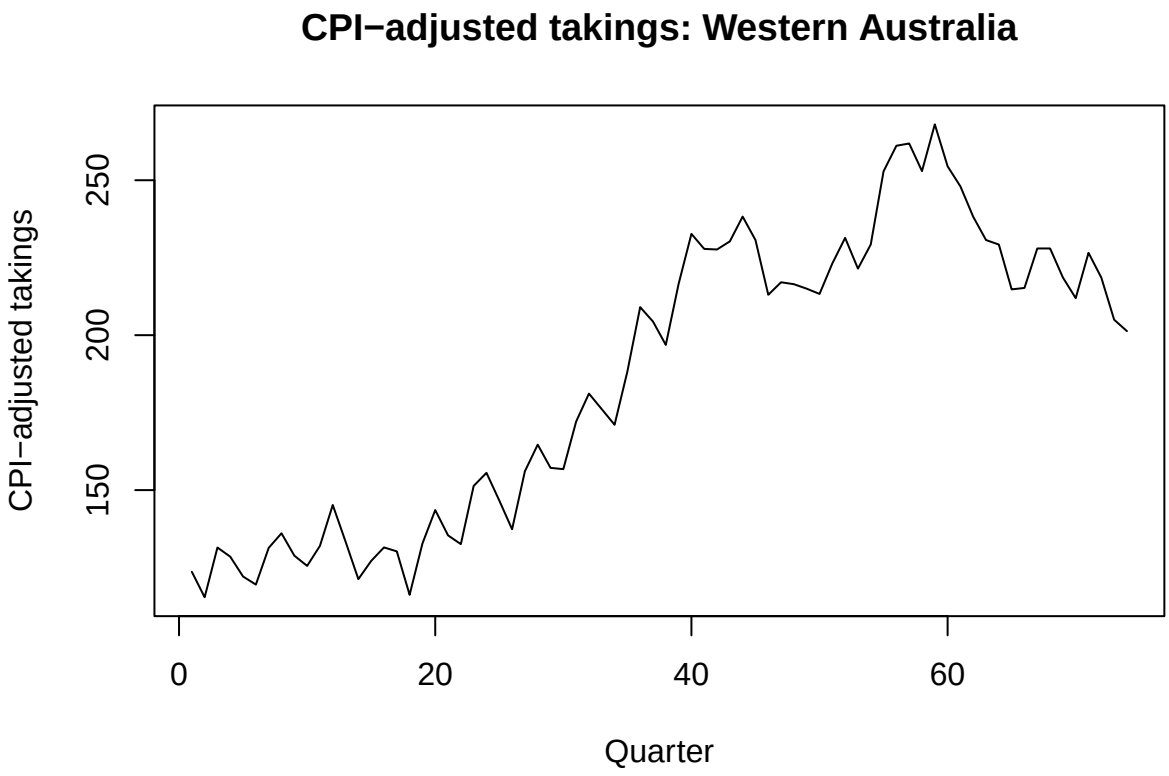

```
tas <- aus_accommodation %>% filter(State == "Tasmania")
plot(tas$cpi_adj_takings, type = "l", main = "CPI-adjusted takings: Tasmania", xlab = "Quarter", ylab =
```




```
vic <- aus_accommodation %>% filter(State == "Victoria")
plot(vic$cpi_adj_takings, type = "l", main = "CPI-adjusted takings: Victoria", xlab = "Quarter", ylab =
```




```
wa <- aus_accommodation %>% filter(State == "Western Australia")
plot(wa$cpi_adj_takings, type = "l", main = "CPI-adjusted takings: Western Australia", xlab = "Quarter"
```



b)

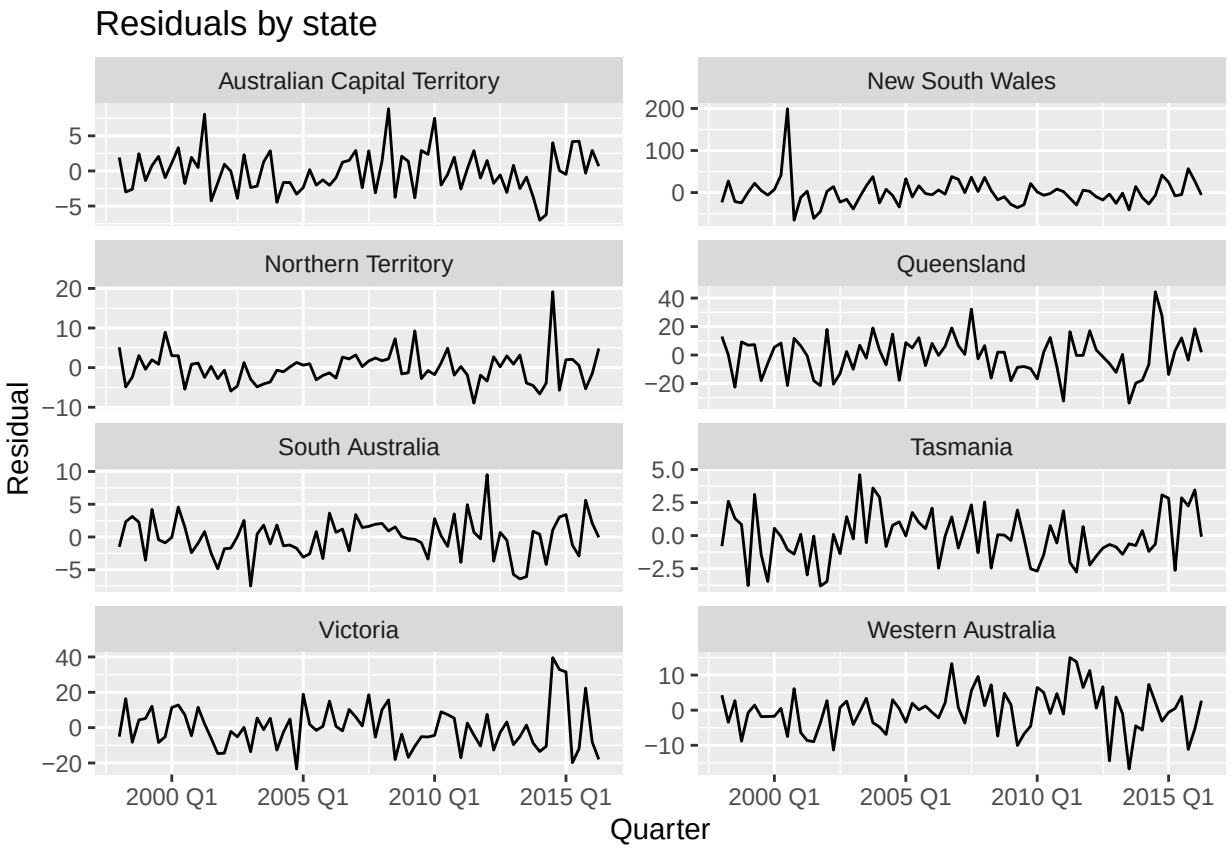
```
idx_nm <- index_var(aus_accommodation)
fit_models <- aus_accommodation %>%
  group_by(State) %>%
  mutate(
    t = row_number(),
    knot_idx = match(yearquarter("2008 Q1"), .data[[idx_nm]]),
    post2008 = pmax(0, t - replace(knot_idx, is.na(knot_idx), 1L))
  ) %>%
  model(
    dyn_reg = ARIMA(cpi_adj_takings ~ t + post2008 + season())
  )
fit_models
```

```
## # A mable: 8 x 2
## # Key:     State [8]
##   State                                                    dyn_reg
##   <chr>                                                    <model>
## 1 Australian Capital Territory          <LM w/ ARIMA(1,0,0) errors>
## 2 New South Wales              <LM w/ ARIMA(1,0,0)(0,0,1)[4] errors>
## 3 Northern Territory           <LM w/ ARIMA(0,0,1)(1,0,0)[4] errors>
## 4 Queensland                   <LM w/ ARIMA(1,0,0)(0,0,1)[4] errors>
## 5 South Australia              <LM w/ ARIMA(1,0,0)(1,0,0)[4] errors>
## 6 Tasmania                     <LM w/ ARIMA(0,0,1)(1,0,0)[4] errors>
```

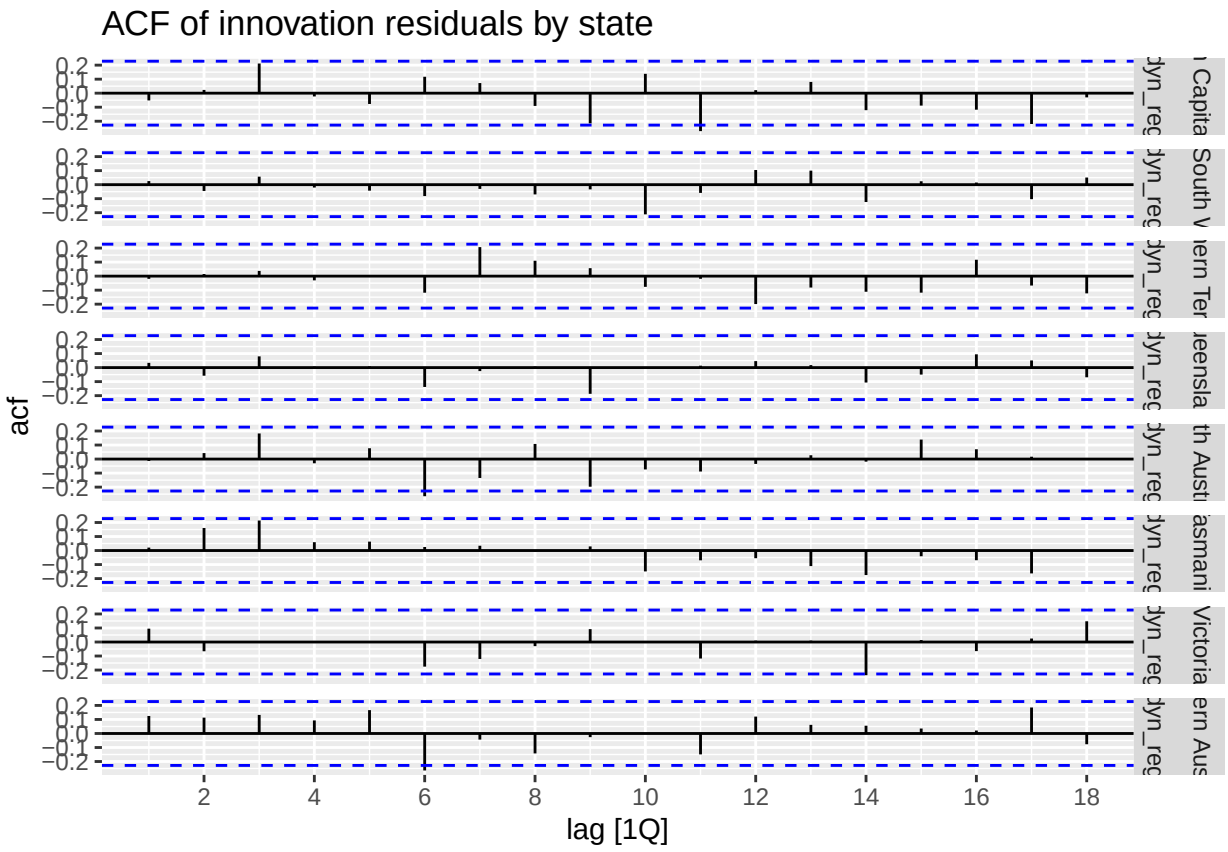
```
## 7 Victoria                    <LM w/ ARIMA(1,0,0)(0,0,1)[4] errors>
## 8 Western Australia                     <LM w/ ARIMA(1,0,0) errors>
```

c)

```
# Residuals from each state's model
resid_df <- fit_models %>% augment()
# Residuals over time by state
resid_df %>%
  ggplot(aes(x = !!sym(idx_nm), y = .resid)) +
  geom_line() +
  facet_wrap(~ State, scales = "free_y", ncol = 2) +
  labs(x = "Quarter", y = "Residual", title = "Residuals by state")
```




```
# ACF of residuals by state (gg_tsresiduals() only accepts one model at a time)
resid_df %>%
  ACF(.innov) %>%
  autoplot() +
  labs(title = "ACF of innovation residuals by state")
```



d)

```
h <- 6

# 1. Forecast CPI with ETS (6 steps)
cpi_series <- aus_accommodation %>%
  filter(State == "New South Wales") %>%
  select(!!sym(idx_nm), CPI)
cpi_fc <- cpi_series %>% model(ETS(CPI)) %>% forecast(h = h) %>% as_tibble()
cpi_fc <- cpi_fc %>% select(Quarter = !!sym(idx_nm), CPI = .mean)

# 2. Forecast CPI-adjusted takings (6 steps) - ETS on adjusted series, no regressors
cpi_adj_fit <- aus_accommodation %>% group_by(State) %>% model(ETS(cpi_adj_takings))
cpi_adj_fc <- cpi_adj_fit %>% forecast(h = h)

# 3. Takings forecast = (CPI-adjusted * CPI) / 100
takings_fc <- cpi_adj_fc %>%
  as_tibble() %>%
  select(State, Quarter = !!sym(idx_nm), cpi_adj_takings = .mean) %>%
  left_join(cpi_fc, by = "Quarter") %>%
  mutate(Takings_forecast = (cpi_adj_takings * CPI) / 100)
takings_fc
```

```
## # A tibble: 48 x 5
##    State                      Quarter cpi_adj_takings   CPI Takings_forecast
```

```
##    <chr>                          <qtr>          <dbl> <dbl>           <dbl>
##  1 Australian Capital Territory 2016 Q3          62.6  109.            68.3
##  2 Australian Capital Territory 2016 Q4          63.0  110.            69.1
##  3 Australian Capital Territory 2017 Q1          63.4  110.            70.0
##  4 Australian Capital Territory 2017 Q2          63.9  111.            70.8
##  5 Australian Capital Territory 2017 Q3          64.3  111.            71.7
##  6 Australian Capital Territory 2017 Q4          64.7  112.            72.5
##  7 New South Wales              2016 Q3         795.   109.           868.
##  8 New South Wales              2016 Q4         880.   110.           966.
##  9 New South Wales              2017 Q1         854.   110.           942.
## 10 New South Wales              2017 Q2         739.   111.           819.
## # i 38 more rows
```

e)

Some uncertainty that has not been taken into account is the CPI forecast uncertainty and the uncertainty of the adjusted takings model. Both of these uncertainties combine into the uncertainty of the takings model.

Problem 2

Harmonic regression (us_gasoline): piecewise linear trend + Fourier terms; select by AICc.

```
head(us_gasoline)
```

```
## # A tsibble: 6 x 2 [1W]
##       Week Barrels
##     <week>   <dbl>
## 1 1991 W06    6.62
## 2 1991 W07    6.43
## 3 1991 W08    6.58
## 4 1991 W09    7.22
## 5 1991 W10    6.88
## 6 1991 W11    6.95
```

```
gas <- us_gasoline |>
  mutate(t = row_number())

knot_vals <- seq(150, nrow(gas) - 150, by = 52)

results <- crossing(knot = knot_vals, K = 1:4) |>
  mutate(
    fit = map2(knot, K, \(k, j) {
      gas |>
        mutate(knot_term = pmax(0, t - k)) |>
        model(
          TSLM(Barrels ~ t + knot_term + fourier(period = 52, K = j))
        )
    }),
    info = map(fit, glance)
  ) |>
  unnest(info) |>
  arrange(AICc)

results |>
```

```
  select(knot, K, AICc) |>
  slice_head(n = 10)
```

```
## # A tibble: 10 x 3
##     knot     K   AICc
##    <dbl> <int>  <dbl>
##  1   774     4 -3356.
##  2   722     4 -3356.
##  3   774     3 -3345.
##  4   722     3 -3344.
##  5   670     4 -3306.
##  6   826     4 -3305.
##  7   670     3 -3295.
##  8   826     3 -3294.
##  9   774     2 -3284.
## 10   722     2 -3284.
```

The best model is a knot at week 774 and K=4.

b)

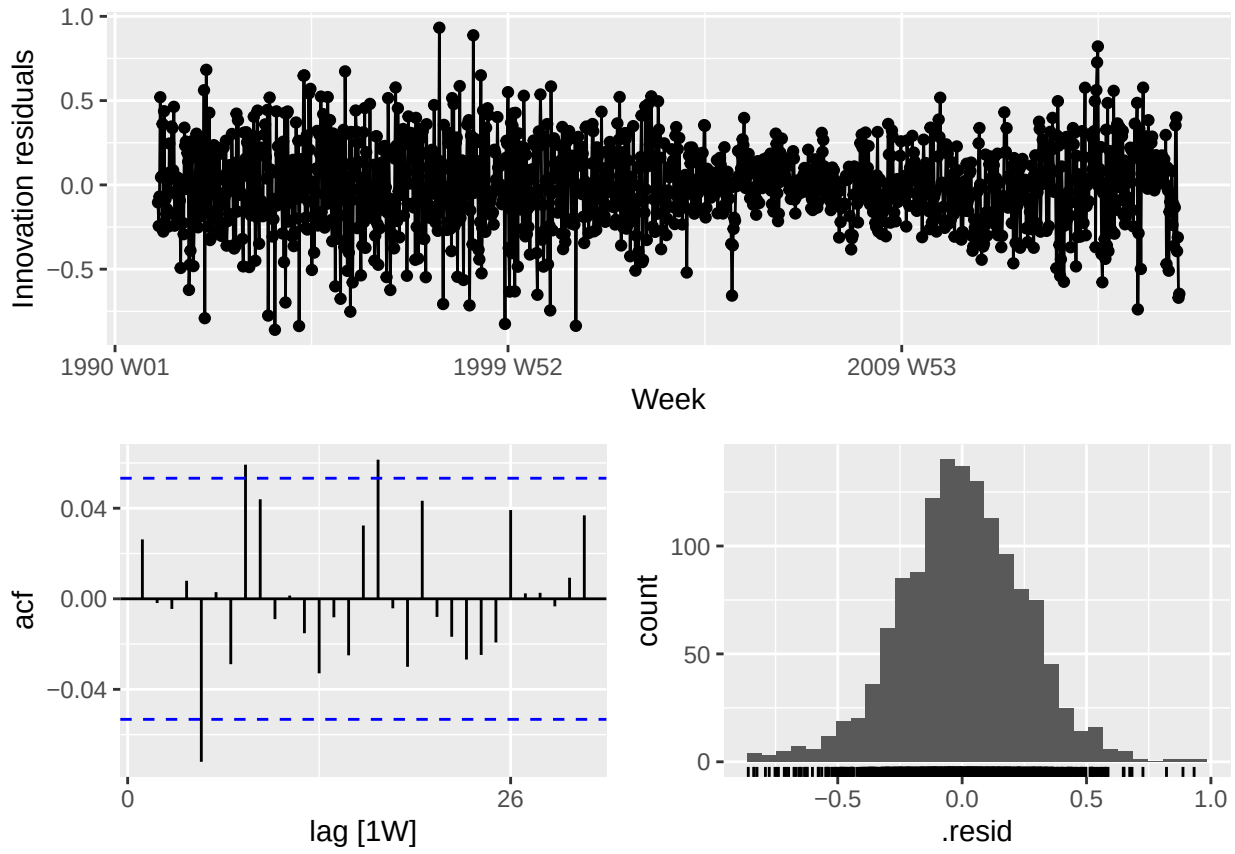
```
best_knot <- 774
best_K <- 4
final_fit <- gas |>
  mutate(knot_term = pmax(0, t - best_knot)) |>
  model(
    ARIMA(
      Barrels ~ t + knot_term + fourier(period = 52, K = best_K) + PDQ(0,0,0),
      stepwise = TRUE,
      approximation = TRUE
    )
  )

glance(final_fit)
```

```
## # A tibble: 1 x 8
##   .model                       sigma2 log_lik   AIC  AICc   BIC ar_roots ma_roots
##   <chr>                         <dbl>   <dbl> <dbl> <dbl> <dbl> <list>   <list>
## 1 "ARIMA(Barrels ~ t + knot_~ 0.0634   -47.6  123.  123.  196. <cpl>    <cpl>
```

c)

```
final_fit |> gg_tsresiduals()
```




```
augment(final_fit) |>
  features(.innov, ljung_box, lag = 52, dof = 10)
```

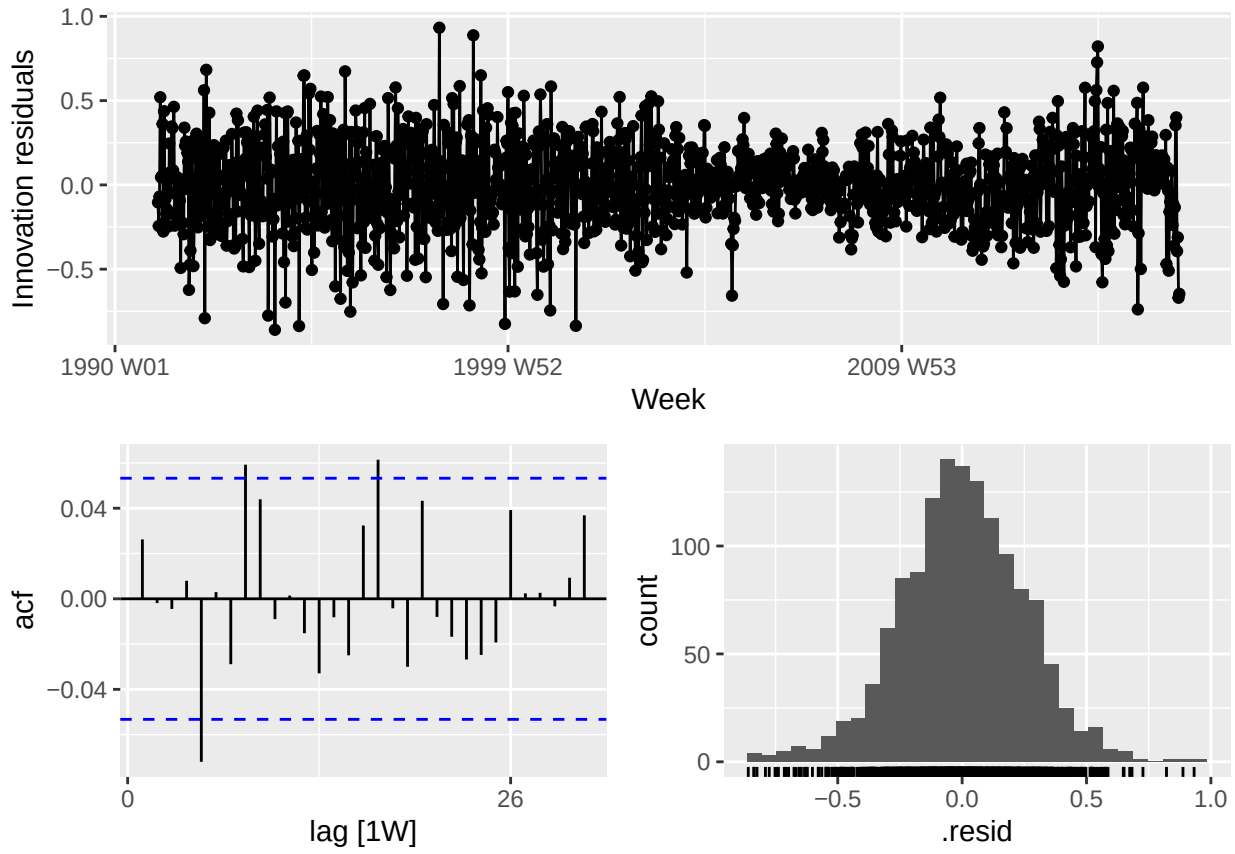
```
## # A tibble: 1 x 3
##   .model                                                  lb_stat lb_pvalue
##   <chr>                                                     <dbl>     <dbl>
## 1 "ARIMA(Barrels ~ t + knot_term + fourier(period = 52, K = b~    103.   4.41e-7
```

```
res <- augment(final_fit)$.resid
Box.test(res, type = "Ljung-Box", lag = 24)
```

```
##
##  Box-Ljung test
##
## data:  res
## X-squared = 32.342, df = 24, p-value = 0.1187
```

d)

```
final_fit |> gg_tsresiduals()
```



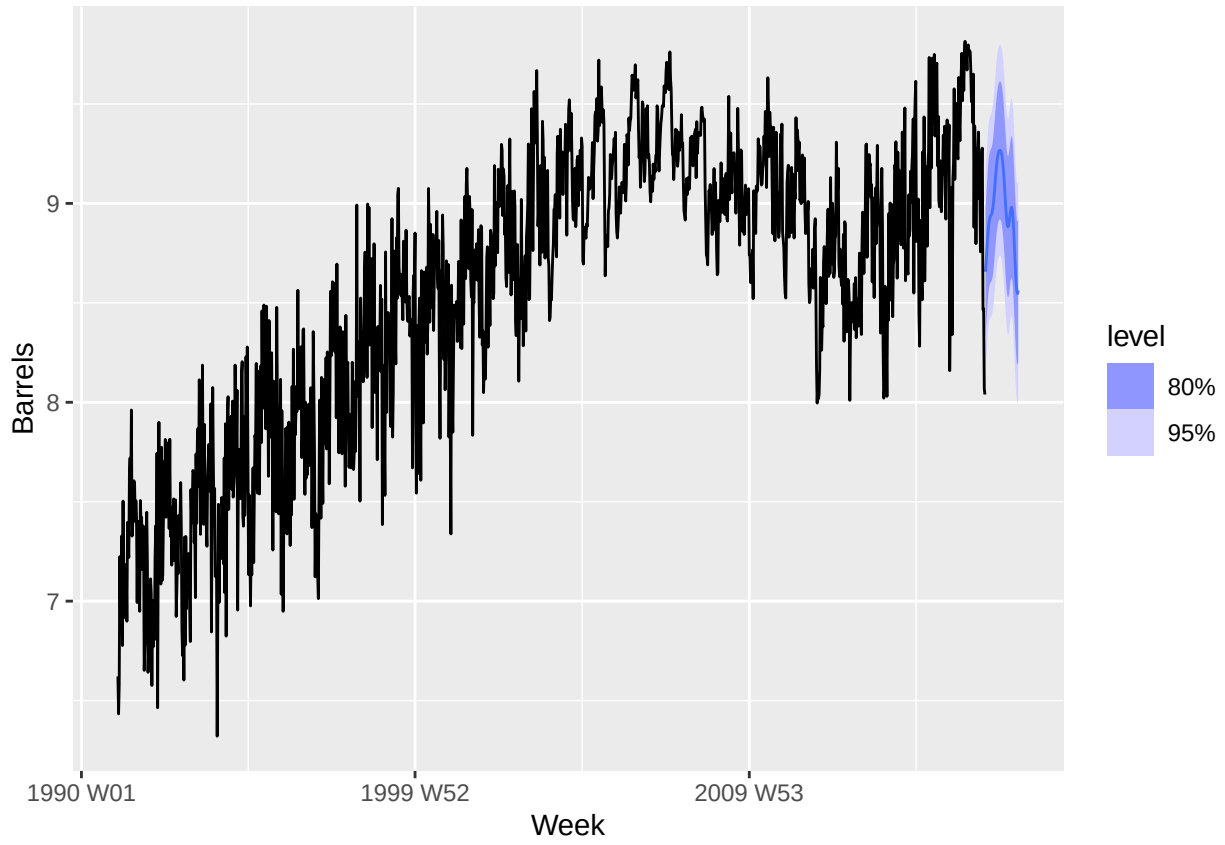

```
augment(final_fit) |>
  features(.innov, ljung_box, lag = 52, dof = 2 * best_K + 2)
```

```
## # A tibble: 1 x 3
##   .model                                                    lb_stat lb_pvalue
##   <chr>                                                       <dbl>     <dbl>
## 1 "ARIMA(Barrels ~ t + knot_term + fourier(period = 52, K = b~    103.   4.41e-7
```

e)

```
# Build future 52 weeks with t and knot_term
n <- nrow(gas)
new_data <- gas |>
  new_data(n = 52) |>
  mutate(t = n + row_number(), knot_term = pmax(0, t - best_knot))

fc <- final_fit |> forecast(new_data = new_data)
fc |> autoplot(gas)
```



Problem 3

a)

This model is an Arima Model with an p=0, d=1, q=1, P=2, D=1, Q=0 where the seasonal frequency is 12.

b)

Since y* is the log of electricity used and x1* and x2* are the square roots of heating and cooling degrees, the coefficients are semi-elasticities.

B1 shows that as the square root of heating degrees increases by one, electricity used increases by about .77% per month on average.

B2 shows that as the square root of cooling degrees increases by one, electricity used increases by about 2.08% per month on average.

c)

Plug in yt* - 0.0077x1* - 0.0208x2* for the error term in the ARIMA. (1+0.5373B12+0.4667B24)(1-B)(1-B12)(yt-*0.0077x1*-0.0208x2*)=(1-0.5830B)et

d)

First forecast the yt* using the beta values. Then, append the arima model as the error term onto this forecast to forecast the full model.

e)

ARIMA is needed because the data is not independent of time and therefore that needs to be accounted for in the model somewhere, arima can do this. Since the data is not independent of time, the statistical tests are not valid if the errors are treated as white noise, but by accounting for the autocorrelation using arima, you can make more valid tests. Modeling the error term (eta_t) as an ARIMA allows for time variation in the data, which is more accurate to the true data generating process and will enhance the forecast.

Problem 4

```
head(aus_retail)
```

```
## # A tsibble: 6 x 5 [1M]
## # Key:       State, Industry [1]
##   State                          Industry             'Series ID'    Month Turnover
##   <chr>                          <chr>                <chr>          <mth>    <dbl>
## 1 Australian Capital Territory Cafes, restaurants~ A3349849A   1982 Apr      4.4
## 2 Australian Capital Territory Cafes, restaurants~ A3349849A   1982 May      3.4
## 3 Australian Capital Territory Cafes, restaurants~ A3349849A   1982 Jun      3.6
## 4 Australian Capital Territory Cafes, restaurants~ A3349849A   1982 Jul      4
## 5 Australian Capital Territory Cafes, restaurants~ A3349849A   1982 Aug      3.6
## 6 Australian Capital Territory Cafes, restaurants~ A3349849A   1982 Sep      4.2
```

```
retail <- aus_retail |> filter(`Series ID` == "A3349849A")
lambda <- retail %>%
  pull(Turnover) %>%
  forecast::BoxCox.lambda()

fit <- model(retail,
  `K = 1` = ARIMA(box_cox(Turnover, lambda) ~ fourier(K = 1) + PDQ(0,0,0)),
  `K = 2` = ARIMA(box_cox(Turnover, lambda) ~ fourier(K = 2) + PDQ(0,0,0)),
  `K = 3` = ARIMA(box_cox(Turnover, lambda) ~ fourier(K = 3) + PDQ(0,0,0)),
  `K = 4` = ARIMA(box_cox(Turnover, lambda) ~ fourier(K = 4) + PDQ(0,0,0)),
  `K = 5` = ARIMA(box_cox(Turnover, lambda) ~ fourier(K = 5) + PDQ(0,0,0)),
  `K = 6` = ARIMA(box_cox(Turnover, lambda) ~ fourier(K = 6) + PDQ(0,0,0)),
  `K = 7` = ARIMA(box_cox(Turnover, lambda) ~ fourier(K = 7) + PDQ(0,0,0)),
  `K = 8` = ARIMA(box_cox(Turnover, lambda) ~ fourier(K = 8) + PDQ(0,0,0)),
  `K = 9` = ARIMA(box_cox(Turnover, lambda) ~ fourier(K = 9) + PDQ(0,0,0)),
  `K = 10` = ARIMA(box_cox(Turnover, lambda) ~ fourier(K = 10) + PDQ(0,0,0))
)

glance(fit) |> arrange(AICc)
```

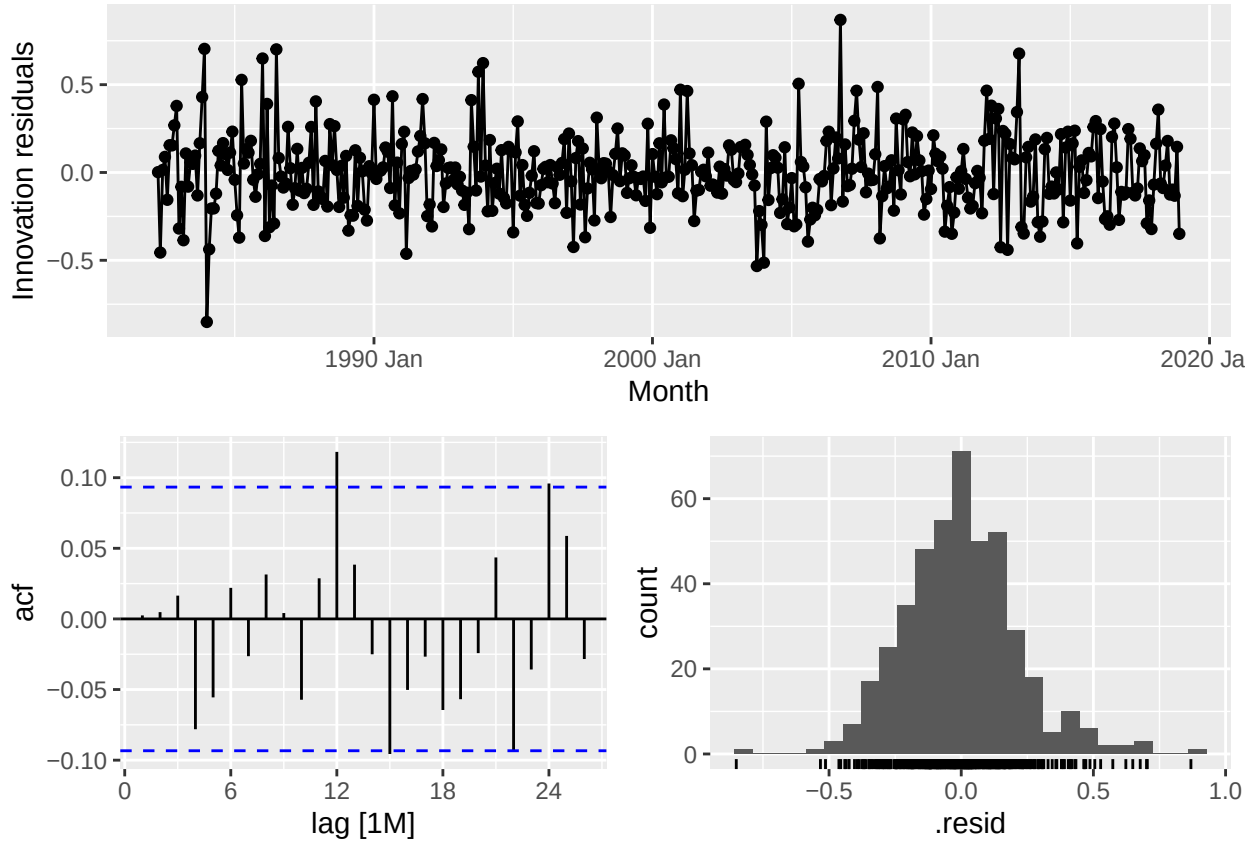
```
## # A tibble: 6 x 10
##   State       Industry .model sigma2 log_lik   AIC  AICc   BIC ar_roots ma_roots
##   <chr>       <chr>    <chr>   <dbl>   <dbl> <dbl> <dbl> <dbl> <list>   <list>
## 1 Australian~ Cafes, ~ K = 6  0.0478   51.5  -72.9 -71.8 -11.6 <cpl>    <cpl>
## 2 Australian~ Cafes, ~ K = 5  0.0481   49.6  -71.2 -70.2 -14.0 <cpl>    <cpl>
## 3 Australian~ Cafes, ~ K = 4  0.0543   22.6  -19.2 -18.4  33.9 <cpl>    <cpl>
## 4 Australian~ Cafes, ~ K = 3  0.0584    5.37  11.3  11.9  56.2 <cpl>    <cpl>
## 5 Australian~ Cafes, ~ K = 2  0.0665  -23.6   67.2  67.7 108.  <cpl>    <cpl>
## 6 Australian~ Cafes, ~ K = 1  0.0737  -47.7  109.  110.  138.  <cpl>    <cpl>
```

```
final_fit <- model(retail,
  ARIMA(box_cox(Turnover, lambda) ~ fourier(K = 6) + PDQ(0,0,0))
)
```

Best K is = 6.

b)

```
final_fit |> gg_tsresiduals()
```



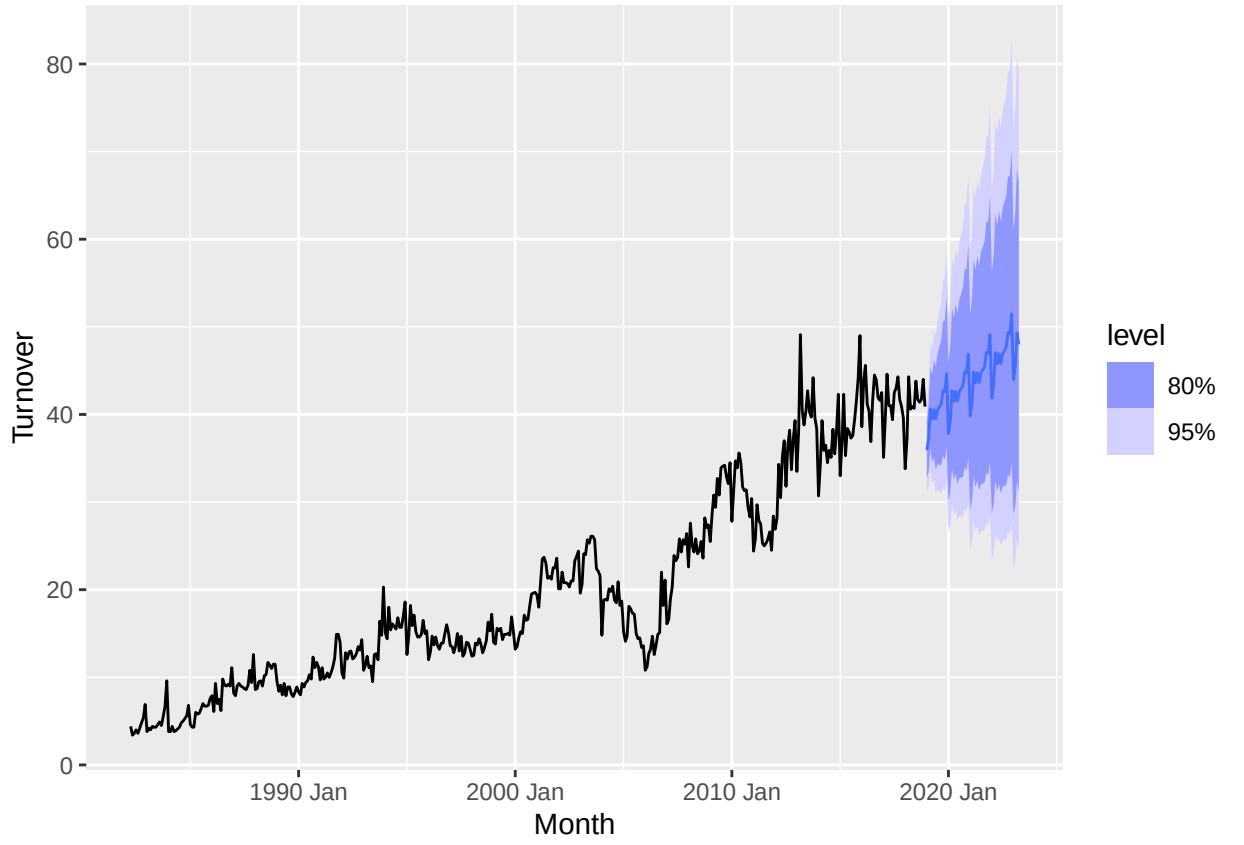

```
augment(final_fit) |>
  features(.innov, ljung_box, lag = 24, dof = 8)
```

```
## # A tibble: 1 x 5
##   State                          Industry                 .model lb_stat lb_pvalue
##   <chr>                          <chr>                    <chr>    <dbl>     <dbl>
## 1 Australian Capital Territory Cafes, restaurants and ~ "ARIM~    33.7   0.00595
```

The residual plots look close to white noise, but the Ljung-Box test has a p-value of .006, so we reject the null hypothesis that the residuals are white noise.

c)

```
forecast(final_fit, h = 52) |> autoplot(retail)
```



It looks similar to forecasts from previous models, however, it does look stronger and captures the seasonal component better while also using less parameters.

Problem 5

```
head(pedestrian)
```

```
## # A tsibble: 6 x 5 [1h] <Australia/Melbourne>
## # Key:       Sensor [1]
##   Sensor          Date_Time           Date        Time Count
##   <chr>           <dttm>              <date>     <int> <int>
## 1 Birrarung Marr 2015-01-01 00:00:00 2015-01-01     0  1630
## 2 Birrarung Marr 2015-01-01 01:00:00 2015-01-01     1   826
## 3 Birrarung Marr 2015-01-01 02:00:00 2015-01-01     2   567
## 4 Birrarung Marr 2015-01-01 03:00:00 2015-01-01     3   264
## 5 Birrarung Marr 2015-01-01 04:00:00 2015-01-01     4   139
## 6 Birrarung Marr 2015-01-01 05:00:00 2015-01-01     5    77
```

```
ped <- pedestrian |>
  filter(Sensor == "Birrarung Marr") |>
  select(Date_Time, Count) |>
  as_tsibble(index = Date_Time) |>
  fill_gaps() |>
```

```
  mutate(Count = if_else(is.na(Count), 0, Count))

# STL decomposition model
fit_stl <- ped |>
  model(
    STL_ETS = decomposition_model(
      STL(Count ~ season(window = "periodic")),
      ETS(season_adjust ~ season("N"))
    )
  )

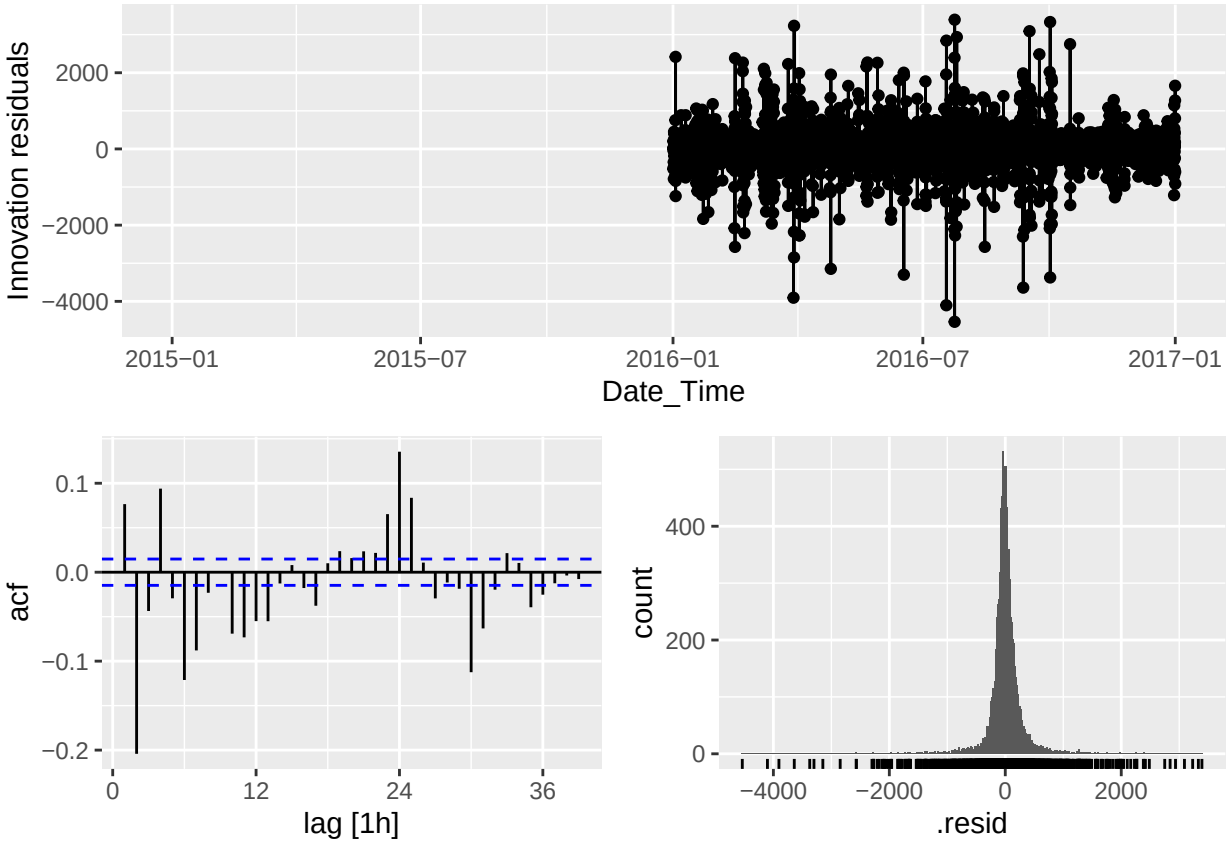
fit_dhr <- ped |>
  model(
    `K = 1` = ARIMA(Count ~ fourier(K = 1) + PDQ(0,0,0)),
    `K = 2` = ARIMA(Count ~ fourier(K = 2) + PDQ(0,0,0)),
    `K = 3` = ARIMA(Count ~ fourier(K = 3) + PDQ(0,0,0)),
    `K = 4` = ARIMA(Count ~ fourier(K = 4) + PDQ(0,0,0)),
    `K = 5` = ARIMA(Count ~ fourier(K = 5) + PDQ(0,0,0)),
    `K = 6` = ARIMA(Count ~ fourier(K = 6) + PDQ(0,0,0))
  )

glance(fit_dhr) |> arrange(AICc)
```

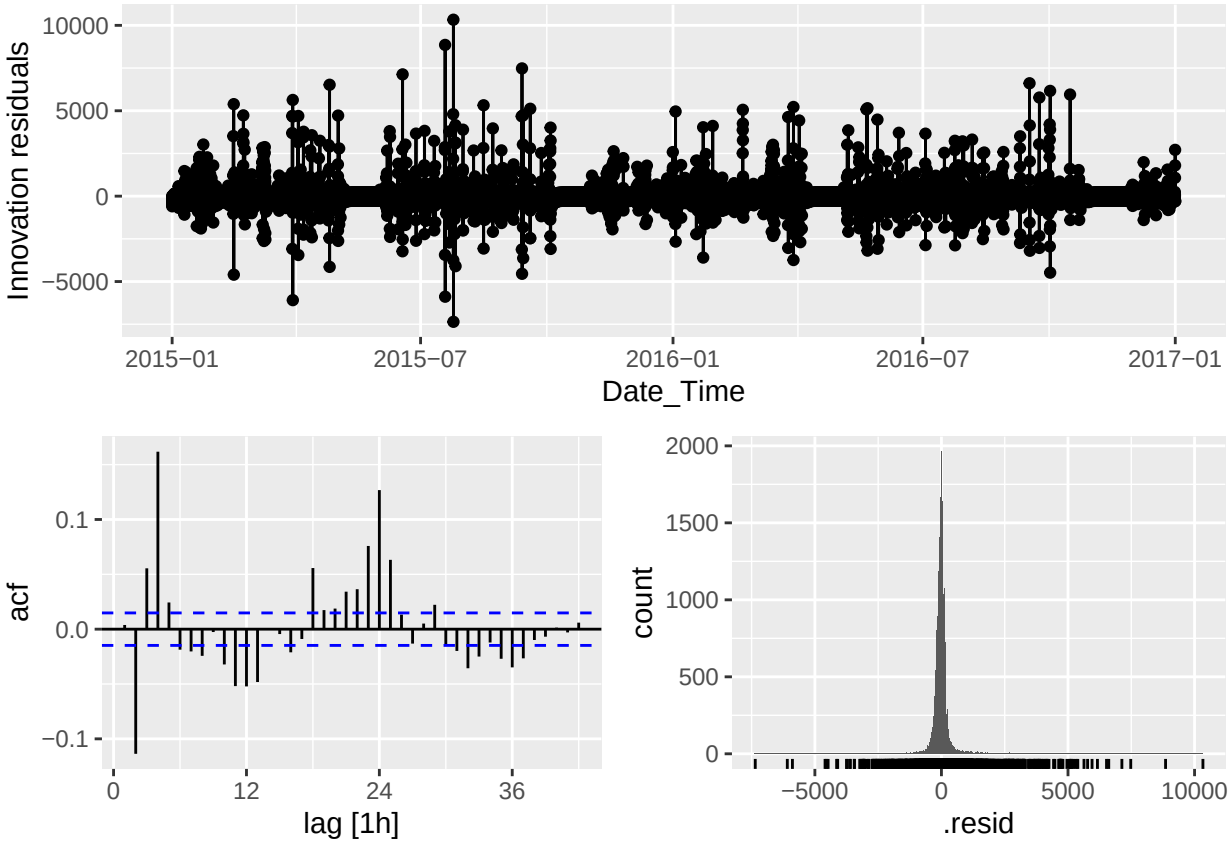
```
## # A tibble: 6 x 8
##   .model  sigma2  log_lik     AIC    AICc     BIC ar_roots  ma_roots
##   <chr>    <dbl>    <dbl>   <dbl>   <dbl>   <dbl> <list>    <list>
## 1 K = 6  237139. -133445. 266923. 266923. 267047. <cpl [1]> <cpl [2]>
## 2 K = 5  237410. -133456. 266941. 266941. 267050. <cpl [1]> <cpl [2]>
## 3 K = 4  242937. -133659. 267342. 267342. 267436. <cpl [1]> <cpl [2]>
## 4 K = 1  243620. -133686. 267387. 267387. 267441. <cpl [2]> <cpl [2]>
## 5 K = 3  243939. -133696. 267413. 267413. 267490. <cpl [1]> <cpl [2]>
## 6 K = 2  244041. -133701. 267418. 267418. 267480. <cpl [1]> <cpl [2]>
```

Best K is = 6.

```
best_K_ped <- 6
final_dhr <- model(ped,
  ARIMA(Count ~ fourier(K = 6) + PDQ(0,0,0))
)
fit_stl |> gg_tsresiduals()
```




```
final_dhr |> gg_tsresiduals()
```




```
augment(fit_stl) |>
  features(.innov, ljung_box, lag = 24)
```

```
## # A tibble: 1 x 3
##   .model  lb_stat lb_pvalue
##   <chr>     <dbl>     <dbl>
## 1 STL_ETS   1097.         0
```

```
augment(final_dhr) |>
  features(.innov, ljung_box, lag = 24, dof = 2 * best_K_ped)
```

```
## # A tibble: 1 x 3
##   .model                                          lb_stat lb_pvalue
##   <chr>                                             <dbl>     <dbl>
## 1 ARIMA(Count ~ fourier(K = 6) + PDQ(0, 0, 0))    1432.         0
```

The two residuals look very similar on the ACF. The Ljung-Box tests have p-values of approximately 0 for both, which shows that neither model captures all of the available information in the data.

c)

The forecast is best from the STL model because its residuals are closer to white noise than the DHR model residuals (a Ljung-Box statistic of 1097 compared to 1432). However, both still have a lot of room for improvement.

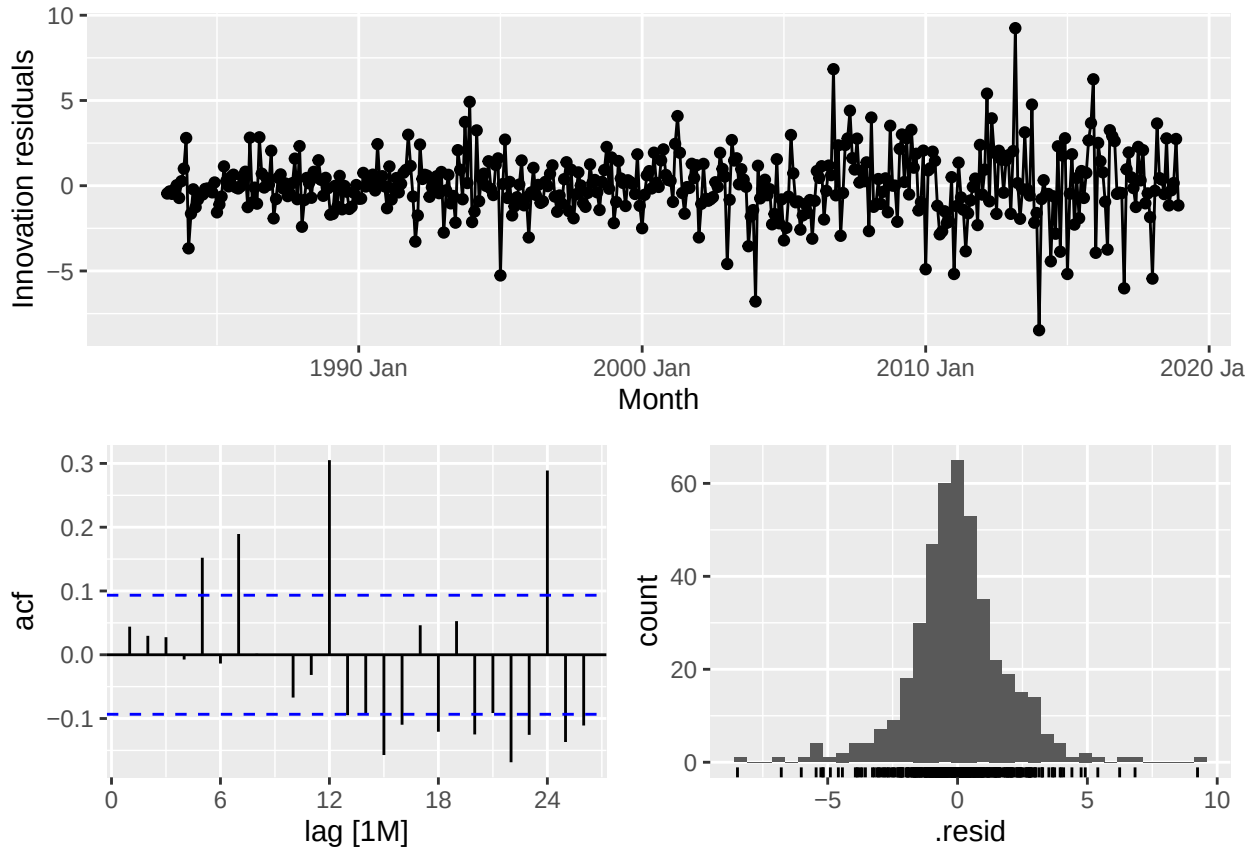
Problem 6

Redundant of earlier problem, Dr. Rojas said we could skip.

Problem 7

```
set.seed(144)
fit_retail <- retail |>
  model(
    NNETAR = NNETAR(Turnover)
  )

gg_tsresiduals(fit_retail)
```




```
# us_gasoline
fit_gas <- us_gasoline |>
  model(
    ARIMA = ARIMA(Barrels),
    ETS = ETS(Barrels),
    NNETAR = NNETAR(Barrels)
  )

fc_gas <- fit_gas |> forecast(h = 12)

accuracy(fit_gas) |> arrange(RMSE)
```

```
## # A tibble: 3 x 10
```

```
##   .model .type          ME  RMSE   MAE     MPE  MAPE  MASE RMSSE     ACF1
##   <chr>  <chr>       <dbl> <dbl> <dbl>   <dbl> <dbl> <dbl> <dbl>    <dbl>
## 1 NNETAR Training 0.000369 0.232 0.180 -0.0833  2.14 0.624 0.638 -0.0298
## 2 ARIMA  Training 0.00358  0.268 0.206 -0.0450  2.48 0.716 0.736 -0.0245
## 3 ETS    Training 0.00378  0.277 0.213 -0.0454  2.56 0.740 0.761 -0.00252
```

```
fc_gas |> autoplot(us_gasoline)
```

